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# ID3 Decision Tree - Employee Promotion

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import pandas as pd
import matplotlib.pyplot as plt
from google.colab import files
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, export_text, plot_tree


# Upload CSV File
print("Upload Employee.csv")
uploaded = files.upload()


# Read Dataset
data = pd.read_csv("Employee.csv")

print("\n===== ORIGINAL DATASET =====\n")
print(data)


# -----
# Label Encoding
# -----


label_encoders = {}


encoded_data = data.copy()


for column in encoded_data.columns:
    le = LabelEncoder()
    encoded_data[column] = le.fit_transform(encoded_data[column])

```

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label_encoders[column] = le

print("\n===== ENCODED DATASET =====\n")
print(encoded_data)

# -----
# Features & Target
# -----

X = encoded_data.drop("Promotion", axis=1)
y = encoded_data["Promotion"]

# -----
# Build ID3 Tree
# -----

model = DecisionTreeClassifier(
    criterion="entropy",
    random_state=42
)

model.fit(X, y)

# -----
# Text Decision Tree
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print("\n===== DECISION TREE RULES =====\n")
print(export_text(model, feature_names=list(X.columns)))

# -----

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# Graphical Decision Tree
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plt.figure(figsize=(20,12))
```

```
plot_tree(  
    model,  
    feature_names=X.columns,  
    class_names=label_encoders["Promotion"].classes_,  
    filled=True,  
    rounded=True,  
    fontsize=10  
)
```

```
plt.title("Employee Promotion Prediction using ID3", fontsize=18)
```

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plt.show()
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# -----
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# New Employee Prediction
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# -----
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```
new_employee = pd.DataFrame({  
    "Experience":["High"],  
    "Performance":["Good"],  
    "Attendance":["Good"],  
    "Training":["Yes"],  
    "Leadership":["Yes"]  
})
```

```
# Encode Input
```

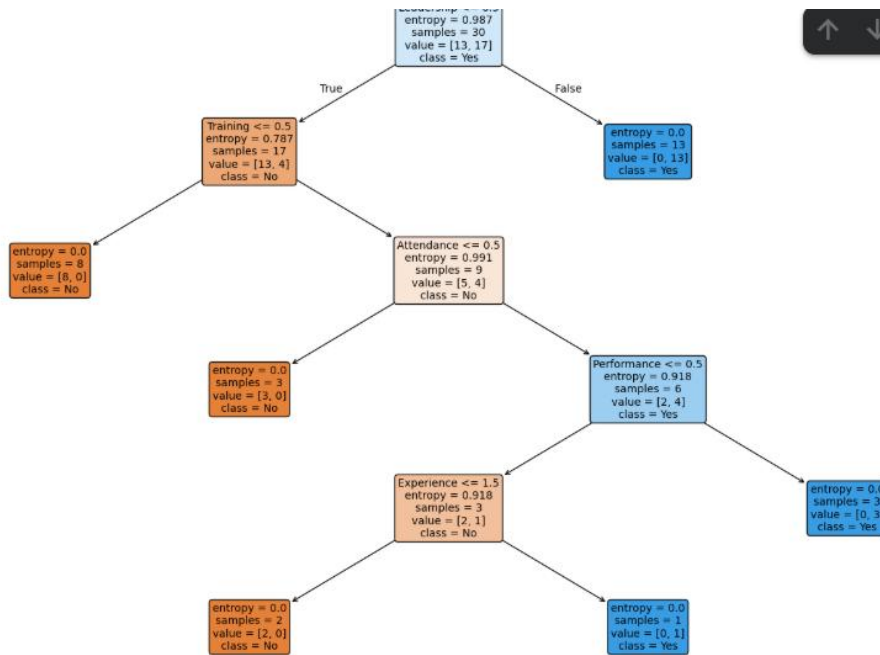
```
for column in new_employee.columns:
    new_employee[column] = label_encoders[column].transform(new_employee[column])

prediction = model.predict(new_employee)

result = label_encoders["Promotion"].inverse_transform(prediction)

print("\n===== NEW EMPLOYEE =====")
print(new_employee)

print("\nPrediction")
print("-----")
print("Promotion :", result[0])
```



===== DECISION TREE RULES =====

```

|--- Leadership <= 0.50
|   |--- Training <= 0.50
|   |   |--- class: 0
|   |--- Training > 0.50
|   |   |--- Attendance <= 0.50
|   |   |   |--- class: 0
|   |   |--- Attendance > 0.50
|   |   |   |--- Performance <= 0.50
|   |   |   |   |--- Experience <= 1.50
|   |   |   |   |   |--- class: 0
|   |   |   |   |--- Experience > 1.50
|   |   |   |   |   |--- class: 1
|   |   |   |--- Performance > 0.50
|   |   |   |--- class: 1
|--- Leadership > 0.50
|   |--- class: 1
  
```